Notes

Importance of Practice for JEE Mains

Introduction

Good morning everyone! The instructor emphasizes the importance of solving assignment questions and practicing regularly for JEE Mains.

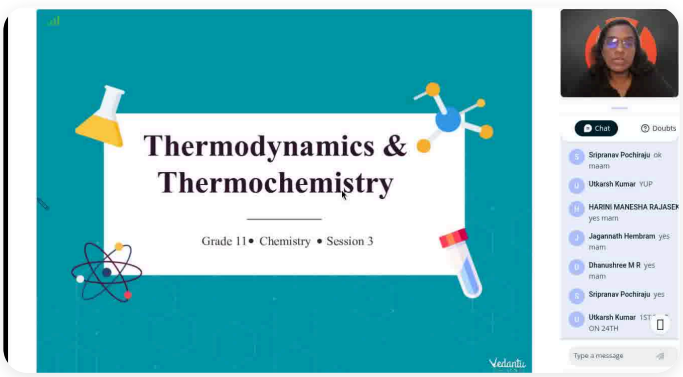
Homework Discussion

- Students are encouraged to bring their doubts regarding assignment questions to the session.
- The instructor will not explain all questions due to time constraints, as it would hinder course completion.
- Students are expected to work through the problems and ask specific questions about numbers they found challenging.

Key to Success

- The most crucial pathway to success in JEE Mains is solving a maximum number of questions.
- Without consistent practice, studying or repeating for JEE Mains is considered a waste of time.
- It is essential to practice extensively and take numerous tests to prepare effectively.

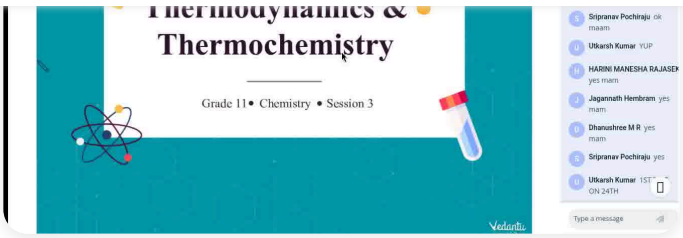
Introduction to Thermodynamics



Overview of Thermodynamics

Today's session will focus on thermodynamics. In the previous class, we covered the calculation of work done for both reversible and irreversible processes. This session will delve into solving questions related to these concepts, and we will also learn how to calculate work done for an adiabatic process and understand heat capacity.

Introduction to Thermodynamics & Thermochemistry



Welcome to Session 3 of Thermodynamics & Thermochemistry

This session is designed for Grade 11 Chemistry students preparing for the JEE exam. We will be reviewing key concepts and solving problems related to the first law of thermodynamics.

Quiz: First Law of Thermodynamics

Question

Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas?

Importance of Revision

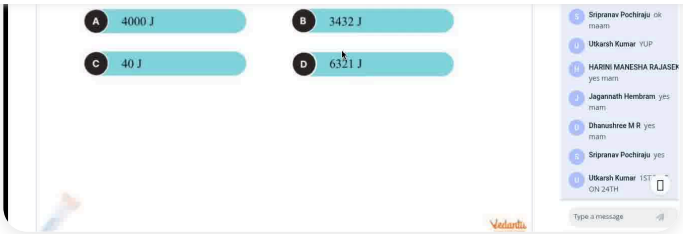
It is crucial to revise the topics taught in previous sessions. Many students struggled with this quiz, indicating a lack of revision. Consistent practice and revision are essential for success in exams like JEE Mains.

Study Strategy for JEE Mains

- **Solve More Questions:** The primary focus for repeaters or students aiming for JEE Mains should be solving a high number of questions daily (e.g., 50-60 questions).
- **Beyond Theory:** While theory is important, simply revising it repeatedly is not enough. After understanding the concepts, the emphasis must shift to applying them through problem-solving.
- **Formulas and Derivations:** Study formulas properly and understand their derivations to effectively solve questions.
- **Active Learning:** Unlike 11th and 12th standard where spoon-feeding might occur, competitive exams require active and independent problem-solving.

Quiz: Work Done in Adiabatic Process

Notes





Question

A real gas is subjected to an adiabatic process from (2 bar, 40 L, 300 K) to (4 bar, 30 L, 300 K) against a constant pressure of 4 bar. Calculate the work done for the process. (Take 1 bar L = 100 J)

A. 4000 J B. 3432 J C. 40 J D. 6321 J

Solution Approach

For a process against a constant external pressure, the work done (W) is given by: $W = -P_{ext}\Delta V$
Where:

- P_{ext} is the constant external pressure.
- ΔV is the change in volume ($V_2 - V_1$).

Given values:

- P_{ext} = 4 bar
- V_1 = 40 L
- V_2 = 30 L

Calculate ΔV: $\Delta V = V_2 - V_1$ = 30 L − 40 L = −10 L

Substitute the values into the work done equation: W = −(4 bar) × (−10 L) W = 40 bar L

Convert bar L to Joules using the given conversion factor (1 bar L = 100 J): $W = 40 \text{ bar L} \times \frac{100 \text{ J}}{1 \text{ bar L}}$
W = 4000 J

Therefore, the work done for the process is 4000 J.

Answer

The correct option is **A. 4000 J**.

First Law of Thermodynamics and Process Types

First Law of Thermodynamics

$\Delta U = Q + W$

Where:

- ΔU is the change in internal energy.
- Q is the heat exchanged with the surroundings.
- W is the work done on the system.

Special Cases of the First Law for Different Processes

1. **Adiabatic Process:**
   - An adiabatic process is one where no heat is exchanged with the surroundings. Therefore, $Q = 0$.
   - The first law simplifies to: $\Delta U = W$.
   - This means any change in internal energy is solely due to work done on or by the system.
2. **Cyclic Process:**
   - In a cyclic process, the system returns to its initial state. Since internal energy is a state function, its change over a complete cycle is zero. Therefore, $\Delta U = 0$.
   - The first law simplifies to: $0 = Q + W$, which means $Q = -W$.
   - This implies that the net heat absorbed by the system is equal to the negative of the net work done on the system (or equal to the net work done *by* the system).
3. **Isochoric Process:**
   - An isochoric process is one where the volume of the system remains constant. Therefore, the change in volume (ΔV) is zero.
   - Work done (W) is given by $W = -P\Delta V$. Since $\Delta V = 0$, the work done is $W = 0$.
   - The first law simplifies to: $\Delta U = Q$.
   - This means that all heat added to the system goes into increasing its internal energy.
4. **Isothermal Process:**
   - An isothermal process is one where the temperature of the system remains constant. For an ideal gas, internal energy is solely a function of temperature. Therefore, for an ideal gas undergoing an isothermal process, the change in internal energy (ΔU) is zero.
   - The first law simplifies to: $0 = Q + W$, which means $Q = -W$.
   - This implies that any heat absorbed by the system is entirely converted into work done by the system, and vice-versa.

Notes

- **Cyclic Process:** $\Delta U = 0 \implies Q = -W$. This statement is **correct**.
- **Adiabatic Process:** $Q = 0 \implies \Delta U = W$. This statement is **correct**.
- **Isochoric Process:** $W = 0 \implies \Delta U = Q$. This statement is **correct**.
- **Isothermal Process:** $\Delta U = 0 \implies Q = -W$. This statement is **correct**.

The question asks which equation *does not correctly represent* the first law. The provided options in the lecture were likely presented in a multiple-choice format, and the lecturer identified one as incorrect. Based on the lecturer's explanation, the statement "For an adiabatic process, when $Q = 0$, $\Delta U = W$" was identified as a wrong statement in the context of the options provided, implying that one of the options might have incorrectly stated $\Delta U = -W$ or some other variation for an adiabatic process. However, based on the fundamental definition, $\Delta U = W$ for an adiabatic process is correct. It's possible the lecturer was referring to a specific option that was *not* explicitly stated in the transcript but was part of the original question. Assuming the lecturer's final conclusion, the incorrect representation would be one that deviates from the derived simplified forms of the first law for each process type.

Work Done in Reversible Expansion of Ideal Gas

Problem Statement

Calculate the work done in ergs for the reversible expansion of one mole of an ideal gas from a volume of 10 liters to 20 liters at 25 degrees Celsius.

Key Concepts and Formulas

- **Reversible Isothermal Expansion Work:** For a reversible isothermal expansion of an ideal gas, the work done (W) is given by the formula: $W = -2.303nRT \log\left(\frac{V_f}{V_i}\right)$ Where:
  - n = number of moles of gas
  - R = ideal gas constant
  - T = temperature in Kelvin
  - V_f = final volume
  - V_i = initial volume

Unit Conversions

It is crucial to use consistent units, especially when the final answer is required in ergs.

- **Energy Units:**
  - 1 Joule (J) = 10^7 ergs
  - 1 Liter-atmosphere (L atm) = 101.3 J

- The standard value of R is 8.314 J/mol·K.
- To convert R to ergs/mol·K: R = 8.314 J/mol\cdotpK × 10^7 ergs/J = 8.314 × 10^7 ergs/mol\cdotpK

- **Temperature Conversion:**
  - Given temperature = 25 °C
  - Convert to Kelvin: T = 25 + 273 = 298 K

Step-by-Step Solution

1. **Identify Given Values:**
   - n = 1 mole
   - V_i = 10 liters
   - V_f = 20 liters
   - T = 298 K
   - R = 8.314 × 10^7 ergs/mol·K
2. **Substitute Values into the Work Formula:** W = −2.303 × (1 mol) × (8.314 × 10^7 ergs/mol\cdotpK) × (298 K) × log $\left(\frac{20\text{ L}}{10\text{ L}}\right)$
3. **Simplify the Logarithmic Term:** log $\left(\frac{20}{10}\right)$ = log(2)
4. **Calculate the Work Done:** W = −2.303 × 1 × 8.314 × 10^7 × 298 × log(2) Using log(2) ≈ 0.3010 W = −2.303 × 8.314 × 10^7 × 298 × 0.3010 W ≈ −5.57 × 10^{10} ergs

Final Answer

The work done in ergs for the reversible expansion is approximately −5.57 × 10^{10} ergs. The negative sign indicates that work is done *by* the system (gas) on the surroundings during expansion.

Introduction to Heat Capacity

Definition of Heat Capacity

Heat capacity, also known as thermal capacity, is a fundamental physical property of matter. It is defined as the amount of heat required to raise the temperature of a substance by one degree Celsius (or one Kelvin).

- **Importance:** Heat capacity is crucial for understanding concepts like internal energy, enthalpy, and calculating work done in adiabatic processes.

Notes

Formula for Total Heat Capacity

The total heat capacity (C_t) of an object can be calculated using the formula:

$C_t = \frac{Q}{\Delta T}$

Where:

- Q = Amount of heat supplied (in Joules)
- ΔT = Change in temperature (in Kelvin or Celsius)

Units of Heat Capacity

The standard unit for heat capacity is Joules per Kelvin (J/K). This is derived directly from the formula, where heat (Q) is in Joules and temperature change (ΔT) is in Kelvin.

Example Calculation

If an object has a heat capacity of $20 J/K$, and its temperature needs to be raised from $200 K$ to $220 K$:

1. **Calculate the change in temperature (ΔT):** $\Delta T = 220 K - 200 K = 20 K$
2. **Use the heat capacity formula to find the heat absorbed (Q):** $C_t = \frac{Q}{\Delta T}$ $20 J/K = \frac{Q}{20 K}$ $Q = 20 J/K \times 20 K = 400 J$

Therefore, $400 J$ of heat is required to raise the temperature of the object by $20 K$.

Properties of Heat Capacity

- **Extensive Property:** Heat capacity is an extensive property, meaning it depends on the amount of substance present. A larger mass of a substance will require more heat to raise its temperature by the same amount compared to a smaller mass of the same substance.
- **Path Function:** Heat capacity is also a path function, meaning its value depends on the path taken during the process (e.g., constant pressure or constant volume).

Mass Dependence of Total Heat Capacity

The total heat capacity (C_t) is dependent on the mass of the substance. This means that the C_t value for $1 kg$ of iron will be different from the C_t value for $100 g$ of iron, even if the temperature change is the same.

Explanation: Consider the formula $C_t = \frac{Q}{\Delta T}$. If we want to raise the temperature of $1 kg$ of iron by $20 K$ and $100 g$ of iron by $20 K$, the ΔT value is the same ($20 K$). However, the amount of heat (

Specific Heat Capacity

Definition

Specific heat capacity is defined as the amount of heat required to raise the temperature of a unit mass of a substance by 1 degree Celsius (or 1 Kelvin). It is calculated for a per gram basis, not for a larger quantity like 100 grams or 10 kg.

Intensive Property

Since specific heat capacity is calculated per unit mass (per gram), it is an **intensive property**. This means it does not depend on the amount of substance present.

Formula

The formula for specific heat capacity (S) is: $S = \frac{Q}{m \cdot \Delta T}$ Where:

- Q = Heat required
- m = Mass of the substance
- ΔT = Change in temperature (in degrees Celsius or Kelvin)

Units

The unit of specific heat capacity is typically **joule per gram per Kelvin** ($J \cdot g^{-1} \cdot K^{-1}$) or **joule per gram per degree Celsius** ($J \cdot g^{-1} \cdot °C^{-1}$).

State Function

Specific heat capacity is a **state function**, meaning its value depends only on the initial and final states of the system, not on the path taken to reach those states. It is not a path function.

Molar Heat Capacity

Definition

Molar heat capacity (C_m) is defined as the amount of heat required to raise the temperature of **one mole** of a substance by 1 degree Celsius (or 1 Kelvin). It is calculated on a per mole basis.

Formula

The formula for molar heat capacity (C_m) is: $C_m = \frac{Q}{n \cdot \Delta T}$ Where:

- Q = Heat required

Notes

Relationship between Heat Capacities

We have discussed three types of heat capacities:

1. **Total Heat Capacity (C):** $C = \frac{Q}{\Delta T}$
2. **Specific Heat Capacity (S):** $S = \frac{Q}{m \cdot \Delta T}$
3. **Molar Heat Capacity (C_m):** $C_m = \frac{Q}{n \cdot \Delta T}$

From these definitions, we can establish relationships:

- Specific heat capacity can be seen as total heat capacity divided by mass: $S = \frac{C}{m}$
- Molar heat capacity can be seen as total heat capacity divided by the number of moles: $C_m = \frac{C}{n}$

Conversely, the total heat capacity can be expressed using specific or molar heat capacity:

- Total Heat Capacity = Specific Heat Capacity × Mass $C = S \cdot m$
- Total Heat Capacity = Molar Heat Capacity × Number of Moles $C = C_m \cdot n$

Specific Heat Capacity Calculation

Problem Statement

209 joules is required to raise the temperature of 50 grams of water by 1 Kelvin. What is the specific heat of water?

Solution

To calculate the specific heat capacity of water, we use the formula: $S = \frac{Q}{m \times \Delta T}$ Where:

- S = Specific heat capacity
- Q = Heat required (in Joules)
- m = Mass of the substance (in grams)
- ΔT = Change in temperature (in Kelvin)

Given values:

- Q = 209 joules
- m = 50 grams
- ΔT = 1 Kelvin

Substitute the values into the formula: $S = \frac{209\ \text{J}}{50\ \text{g} \times 1\ \text{K}}$

Notes

Quiz: Heat Required to Raise Temperature

Question

The heat required to raise the temperature of a body by 1 Kelvin is called what?

Explanation

This question tests the understanding of basic thermodynamic definitions. The key is to differentiate between heat capacity and specific heat capacity.

- **Heat Capacity:** The amount of heat energy required to raise the temperature of an entire body or substance by 1 Kelvin (or 1 degree Celsius). It does not specify the mass of the substance.
- **Specific Heat Capacity:** The amount of heat energy required to raise the temperature of *one unit mass* (e.g., one gram or one kilogram) of a substance by 1 Kelvin (or 1 degree Celsius).

Since the question asks for the heat required to raise the temperature of "a body" by 1 Kelvin without specifying its mass, the correct answer is heat capacity.

Answer

Heat Capacity

Calculating Heat Required Using Molar Heat Capacity

Problem Statement

Calculate the amount of heat required to raise the temperature of 60 grams of aluminum from 35 degrees Celsius to 55 degrees Celsius. The molar heat capacity of aluminum is given as 24 J mol^{-1} K^{-1}. The molar mass of aluminum is 27 grams.

Given Data

- Mass of aluminum (m) = 60 g
- Initial temperature (T_1) = 35 degrees Celsius
- Final temperature (T_2) = 55 degrees Celsius
- Molar heat capacity (C_m) = 24 J mol^{-1} K^{-1}
- Molar mass of aluminum (M) = 27 g/mol

Formula for Molar Heat Capacity

- Q is the heat required (in Joules)
- n is the number of moles
- ΔT is the change in temperature

Rearranging the Formula to Find Heat (Q)

We need to calculate the amount of heat (Q), so we can rearrange the formula: $Q = C_m \cdot n \cdot \Delta T$

Step-by-Step Solution

1. **Calculate the change in temperature (ΔT):** $\Delta T = T_2 - T_1$ = 55 degrees Celsius − 35 degrees Celsius = 20 degrees Celsius Since a change of 1 degree Celsius is equal to a change of 1 Kelvin, ΔT = 20 K.

2. **Calculate the number of moles (n) of aluminum:** The number of moles is given by the formula: $n = \frac{\text{given mass}}{\text{molar mass}}$ $n = \frac{60\text{ g}}{27\text{ g/mol}} = \frac{20}{9}$ mol ≈ 2.22 mol

3. **Substitute the values into the rearranged heat formula:** $Q = C_m \cdot n \cdot \Delta T$ $Q = (24\text{ J mol}^{-1}\text{ K}^{-1}) \cdot \left(\frac{20}{9}\text{ mol}\right) \cdot (20\text{ K})$ $Q = 24 \cdot \frac{20}{9} \cdot 20$ J $Q = \frac{24 \cdot 400}{9}$ J $Q = \frac{9600}{9}$ J $Q \approx 1066.67$ J

4. **Convert the heat to kilojoules (kJ):** To convert Joules to kilojoules, divide by 1000: $Q = \frac{1066.67}{1000}$ kJ $Q \approx 1.067$ kJ

Final Answer

The amount of heat required is approximately 1.067 kilojoules.

Relation Between C_P and C_V

Introduction

We will explore the relationship between molar heat capacity at constant pressure (C_P) and molar heat capacity at constant volume (C_V). The fundamental relation is given by: $C_P - C_V = R$ where R is the ideal gas constant.

Molar Heat Capacity

- **Molar heat capacity at constant pressure (C_P)**: This is the heat capacity measured when the pressure is kept constant.
- **Molar heat capacity at constant volume (C_V)**: This is the heat capacity measured when the volume is kept constant.

- It is defined as the heat absorbed at constant pressure (Q_P) per mole (n) per unit change in temperature (ΔT).

$C_{MP} = \frac{Q_P}{n\Delta T}$

- Rearranging this, the heat absorbed at constant pressure is:

$Q_P = nC_{MP}\Delta T$

2. **Molar Heat Capacity at Constant Volume (C_{MV})**:
   - It is defined as the heat absorbed at constant volume (Q_V) per mole (n) per unit change in temperature (ΔT).

$C_{MV} = \frac{Q_V}{n\Delta T}$

- Rearranging this, the heat absorbed at constant volume is:

$Q_V = nC_{MV}\Delta T$

First Law of Thermodynamics and Heat Changes

- According to the First Law of Thermodynamics, $\Delta U = Q + W$.
- **At constant volume**: Work done (W) is zero because $W = -P_{ext}\Delta V$, and if $\Delta V = 0$, then $W = 0$.
  - Therefore, at constant volume, the First Law becomes:

$\Delta U = Q_V$

- Substituting the expression for Q_V:

$\Delta U = nC_{MV}\Delta T$ This equation relates the change in internal energy to the molar heat capacity at constant volume.

- **At constant pressure**: The heat change at constant pressure (Q_P) is defined as the change in enthalpy (ΔH).
  - Therefore:

$\Delta H = Q_P$

- Substituting the expression for Q_P:

$\Delta H = nC_{MP}\Delta T$ This equation relates the change in enthalpy to the molar heat capacity at constant pressure.

- Heat change at constant pressure (Q_P) is equal to the change in enthalpy (ΔH).

These fundamental relationships will be used to derive the relation between C_P and C_V.

Relation between Molar Heat Capacities (C_p and C_v)

Derivation of $C_p - C_v = R$

We begin by considering the first law of thermodynamics, which states: $\Delta U = Q + W$ Where:

- ΔU is the change in internal energy.
- Q is the heat exchanged.
- W is the work done.

For an isobaric process (constant pressure), the heat change (Q_p) is given by: $Q_p = nC_p\Delta T$ And the work done (W) is given by: $W = -P\Delta V$ Substituting these into the first law of thermodynamics: $\Delta U = nC_p\Delta T - P\Delta V$

We also know that for a constant volume process, the change in internal energy is given by: $\Delta U = nC_v\Delta T$

Equating the expressions for ΔU: $nC_v\Delta T = nC_p\Delta T - P\Delta V$

Now, let's express ΔV as $V_2 - V_1$: $nC_v\Delta T = nC_p\Delta T - P(V_2 - V_1)$ $nC_v\Delta T = nC_p\Delta T - (PV_2 - PV_1)$

Using the ideal gas law, $PV = nRT$:

- $PV_2 = nRT_2$
- $PV_1 = nRT_1$

Substitute these into the equation: $nC_v\Delta T = nC_p\Delta T - (nRT_2 - nRT_1)$ $nC_v\Delta T = nC_p\Delta T - nR(T_2 - T_1)$ Since $\Delta T = T_2 - T_1$: $nC_v\Delta T = nC_p\Delta T - nR\Delta T$

Divide the entire equation by $n\Delta T$ (assuming $n \neq 0$ and $\Delta T \neq 0$): $C_v = C_p - R$

Rearranging the terms, we get the final relation: $C_p - C_v = R$

This relation holds for one mole of a substance ($n = 1$). R is the ideal gas constant.

Key Takeaway

- The difference between the molar heat capacity at constant pressure (C_p) and the molar heat capacity at constant volume (C_v) is equal to the ideal gas constant (R).

Poisson's Ratio and Relations between Cp, Cv, and γ

Introduction to Poisson's Ratio

Poisson's ratio, denoted by γ (gamma), is defined as the ratio of the molar specific heat at constant pressure (C_p) to the molar specific heat at constant volume (C_v).

$\gamma = \frac{C_p}{C_v}$

Derivation of C_v in terms of R and γ

We know the relationship between C_p and C_v for an ideal gas:

$C_p - C_v = R$

From this, we can write $C_p = C_v + R$.

Now, substitute this into the definition of γ:

$\gamma = \frac{C_v+R}{C_v}$

Divide each term in the numerator by C_v:

$\gamma = \frac{C_v}{C_v} + \frac{R}{C_v}$

$\gamma = 1 + \frac{R}{C_v}$

To find C_v, rearrange the equation:

$\gamma - 1 = \frac{R}{C_v}$

Therefore, C_v can be expressed as:

$C_v = \frac{R}{\gamma-1}$

Derivation of C_p in terms of R and γ

We already know that $C_p = \gamma C_v$.

Substitute the expression for C_v we just derived:

$C_p = \gamma \left(\frac{R}{\gamma-1}\right)$

Thus, C_p can be expressed as:

$C_p = \frac{\gamma R}{\gamma-1}$

Notes

Introduction

This section discusses the relationship between molar heat capacities at constant pressure (C_p) and constant volume (C_v) for different types of ideal gases (monoatomic, diatomic, and polyatomic). This relationship is crucial for understanding the thermodynamic behavior of gases.

Key Relationship

The fundamental relationship between C_p and C_v for an ideal gas is given by: $C_p - C_v = R$
Where R is the ideal gas constant.

Molar Heat Capacities for Different Gas Types

It is important to memorize the values of C_v and C_p for various types of ideal gases:

- **Monoatomic Gas**
  - Examples: Neon (Ne), Argon (Ar), Helium (He)
  - $C_v = \frac{3}{2}R$
  - Using the relation $C_p - C_v = R$, we can find C_p: $C_p = C_v + R = \frac{3}{2}R + R = \frac{5}{2}R$
- **Diatomic Gas**
  - Examples: Hydrogen (H_2), Oxygen (O_2), Nitrogen (N_2), Chlorine (Cl_2)
  - $C_v = \frac{5}{2}R$
  - Using the relation $C_p - C_v = R$, we can find C_p: $C_p = C_v + R = \frac{5}{2}R + R = \frac{7}{2}R$
- **Non-linear Polyatomic Gas**
  - Examples: Sulfur Trioxide (SO_3), Methane (CH_4), Ammonia (NH_3)
  - $C_v = 3R$
  - Using the relation $C_p - C_v = R$, we can find C_p: $C_p = C_v + R = 3R + R = 4R$

Summary Table

Gas Type	C_v **(Molar Heat Capacity at Constant Volume)**	C_p **(Molar Heat Capacity at Constant Pressure)**
Monoatomic	$\frac{3}{2}R$	$\frac{5}{2}R$
Diatomic	$\frac{5}{2}R$	$\frac{7}{2}R$
Non-linear Polyatomic	$3R$	$4R$

Notes

Introduction

For an ideal gas, the relationship between the molar specific heat at constant pressure (C_p) and the molar specific heat at constant volume (C_v) is crucial for various thermodynamic calculations. We will explore how to determine C_p given C_v for a monoatomic ideal gas.

Key Relations

There are two primary relations that connect C_p and C_v:

1. **Mayer's Relation:** $C_p - C_v = R$
2. **Ratio of Specific Heats:** $C_p/C_v = \gamma$ (gamma)

Problem Statement

For a monoatomic ideal gas, the value of C_v is given as 3 calorie per mole per Kelvin. We need to find the value of C_p for the same gas.

Gas Constant (R) Values

The gas constant R has different values depending on the units used:

- In Joules: R = 8.314 Joule per Kelvin per mole
- In Ergs: R = 8.314×10^7 ergs per Kelvin per mole (since 1 Joule = 10^7 ergs)
- In Liter-Atmosphere: R = 0.0821 liter atmosphere per Kelvin per mole
- In Calories: R = 2 calorie per mole per Kelvin

Solution using Mayer's Relation

Given C_v = 3 calorie per mole per Kelvin. We use Mayer's relation: $C_p - C_v = R$. Rearranging for C_p: $C_p = C_v + R$.

Since C_v is given in calories, we must use the value of R in calories. R = 2 calorie per mole per Kelvin.

Substitute the values: C_p = 3 calorie per mole per Kelvin + 2 calorie per mole per Kelvin C_p = 5 calorie per mole per Kelvin

Therefore, the value of C_p for the monoatomic ideal gas is 5 calorie per mole per Kelvin.

Quiz: Molar Heat Capacity of Water

Problem Statement

Solution

To find the increase in temperature (ΔT), we use the formula for molar heat capacity:

$$Q = n \cdot C_m \cdot \Delta T$$

Where:

- Q is the heat supplied.
- n is the number of moles.
- C_m is the molar heat capacity at constant pressure.
- ΔT is the change in temperature.

We are given:

- Q = 750 J
- n = 1 mole
- C_m = 75 J mol^{-1} K^{-1}

Rearranging the formula to solve for ΔT:

$$\Delta T = \frac{Q}{n \cdot C_m}$$

Substitute the given values into the equation:

$$\Delta T = \frac{750 \text{ J}}{1 \text{ mol} \cdot 75 \text{ J mol}^{-1} \text{ K}^{-1}}$$

$$\Delta T = \frac{750}{75} \text{ K}$$

$$\Delta T = 10 \text{ K}$$

Final Answer

The increase in temperature of water is 10 K.

Problem: Calculating Internal Energy and PV Work for an Ideal Gas

Question

5 moles of an ideal gas at 100 Kelvin are allowed to undergo compression until its temperature becomes 200 Kelvin. If C_v is given, calculate ΔU and $\Delta(PV)$ for this process.

Given Data

- Number of moles (n) = 5 moles

- Molar heat capacity at constant volume (C_v) = 28 J/mol·K
- Gas constant (R) = 8 J/mol·K (implied from solution, as it's a common approximation for calculations)

Part 1: Calculate Change in Internal Energy (ΔU)

The change in internal energy for an ideal gas is given by the formula:

$\Delta U = nC_v\Delta T$

Where:

- n = number of moles
- C_v = molar heat capacity at constant volume
- ΔT = change in temperature ($T_2 - T_1$)

First, calculate ΔT:

$\Delta T = T_2 - T_1$ = 200 K − 100 K = 100 K

Now, substitute the values into the formula for ΔU:

ΔU = (5 mol) × (28 J/mol\cdotpK) × (100 K)

ΔU = 5 × 28 × 100 J

ΔU = 14000 J

To convert Joules to kilojoules, divide by 1000:

$\Delta U = \frac{14000}{1000}$ kJ = 14 kJ

Part 2: Calculate Change in PV ($\Delta(PV)$)

For an ideal gas, the ideal gas law states:

$PV = nRT$

The change in the product PV can be expressed as:

$\Delta(PV) = P_2V_2 - P_1V_1$

Since $PV = nRT$, we can write:

$\Delta(PV) = nRT_2 - nRT_1$

Factor out nR:

Substitute the given values:

$\Delta(PV)$ = (5 mol) × (8 J/mol\cdotpK) × (100 K)

$\Delta(PV)$ = 5 × 8 × 100 J

$\Delta(PV)$ = 4000 J

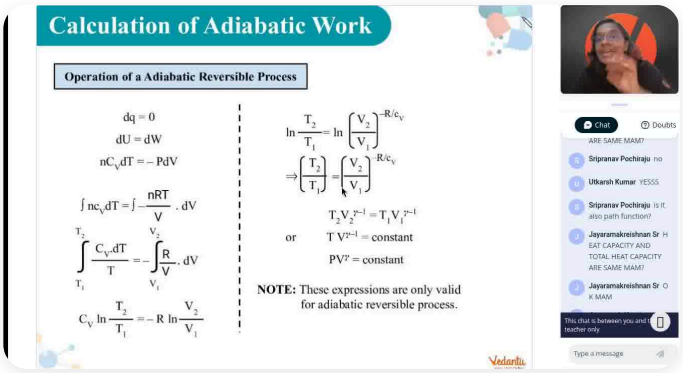
To convert Joules to kilojoules, divide by 1000:

$\Delta(PV) = \frac{4000}{1000}$ kJ = 4 kJ

Final Answer

- ΔU = 14 kJ
- $\Delta(PV)$ = 4 kJ

Work Done in an Adiabatic Process



Introduction

Work done in an adiabatic process is a crucial topic. We will calculate work done for both adiabatic reversible and irreversible processes. Interestingly, the formula for work done in both expansion and compression, whether reversible or irreversible, is the same.

Derivation of Work Done Formula

We start with the first law of thermodynamics: $dU = dQ + dW$ For an adiabatic process, the change in heat (dQ) is zero. Therefore, the equation simplifies to: $dU = dW$ We also know the relation for the change in internal energy (dU) in terms of molar heat capacity at constant volume (C_v) and change in temperature (dT): $dU = nC_v dT$ Substituting this into the simplified first law equation, we get: $dW = nC_v dT$

the ideal gas equation, $PV = nRT$, we can express T as $T = \frac{PV}{nR}$. Substituting this into the equation for work done: $W = n\left(\frac{R}{\gamma-1}\right)\left(\frac{P_2V_2}{nR} - \frac{P_1V_1}{nR}\right)$ The nR terms cancel out: $W = \frac{P_2V_2-P_1V_1}{\gamma-1}$

Final Formula

This formula for work done is applicable for both adiabatic reversible and irreversible processes, whether it's expansion or compression: $W = \frac{P_2V_2-P_1V_1}{\gamma-1}$

Note on Adiabatic Reversible Process

The following expressions are specifically valid for an adiabatic reversible process:

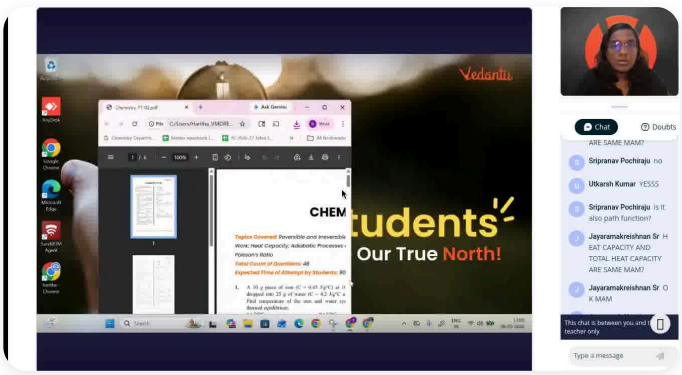
- $dq = 0$
- $dU = dW$
- $nC_v dT = -PdV$
- $TV^{\gamma-1}$ = constant
- PV^{γ} = constant

These expressions are derived from the integration of $nC_v dT = -PdV$ and using the ideal gas law and the relation $C_v \ln \frac{T_2}{T_1} = -R \ln \frac{V_2}{V_1}$.

Polytropic Process

The polytropic process is a concept for JEE Advanced and is generally not asked in JEE Mains.

Assignment Discussion: Overview and Solved Problem



Assignment Overview

This session focuses on discussing questions from the shared assignment, which covers topics such as reversible work, irreversible work, heat capacity, adiabatic processes, and Poisson's ratio. Approximately 90% of these topics have been covered. While adiabatic process graphs and related questions will be addressed in the next class, students are encouraged to start solving questions based on the concepts already taught.

atmospheres to 1 atmosphere at 300 Kelvin. What is the largest mass which can be lifted through a height of 100 meters?

Solution:

1. **Calculate the work done (W) during the isothermal reversible expansion:** For an isothermal reversible process, the work done is given by: $W = -2.303 \cdot nRT \cdot \log\left(\frac{P_{\text{initial}}}{P_{\text{final}}}\right)$ Where:
   - n = 10 moles (number of moles)
   - R = Gas constant (use appropriate value, e.g., 8.314 J $\text{mol}^{-1}\text{K}^{-1}$ or 0.0821 L atm $\text{mol}^{-1}\text{K}^{-1}$ depending on units)
   - T = 300 K (temperature)
   - P_{initial} = 10 atm (initial pressure)
   - P_{final} = 1 atm (final pressure)

   Substitute the values into the formula to calculate W.
2. **Relate work done to potential energy:** The work done by the gas is used to lift a mass against gravity. This work is equal to the gravitational potential energy gained by the mass. $W = mgh$ Where:
   - m = mass to be lifted (in kg)
   - g = 9.8 m/s^2 (acceleration due to gravity)
   - h = 100 m (height)
3. **Solve for mass (m):** Equate the work done from step 1 to the potential energy expression from step 2: $2.303 \cdot nRT \cdot \log\left(\frac{P_{\text{initial}}}{P_{\text{final}}}\right) = mgh$ Rearrange to solve for m: $m = \frac{2.303 \cdot nRT \cdot \log\left(\frac{P_{\text{initial}}}{P_{\text{final}}}\right)}{gh}$

   By substituting the calculated value of W and the given values for g and h, the mass m can be determined. This problem requires applying both thermodynamics and basic physics principles.

Important Announcement and Contact Information

Assignment and Feedback

- Students are encouraged to solve the maximum number of questions from the shared assignment.
- Feedback on the session is highly appreciated at the end.

Personal Telegram ID

Notes

Did you find these notes helpful?